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# Directional solidification microstructures in diffusive and convective regimes

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**Abstract**

Most theoretical models on the development of solidification microstructures are based on diffusive growth conditions, whereas convection effects are generally present in most directional solidification experiments. This paper examines the effect of convection on the development of microstructures in directionally solidified Al–Cu alloys. A numerical model of convection is first discussed to characterize fluid flow as a function of sample diameter for the thermal profile of the directional solidification used in this study, and it was established that sample diameters of 1 mm or smaller is required for diffusive growth. Based on this result, a new experimental design is developed for directionally solidifying several thin samples of different diameters in the same experimental run to systematically study the effect of sample diameter on microstructure development. Sample diameters, ranging from 0.2 to 5.0 mm, were used to access both the diffusive and convective regimes. Experiments have been performed over a range of velocities and temperature gradients to characterize the effect of convection on planar, cellular and dendritic microstructures. In very thin samples of $\leqslant 1.0$ mm diameter, diffusive growth was obtained. The cell/dendrite tip composition and primary spacing were found to increase, and the tip temperature was found to decrease, as the convection effects were reduced in thinner samples. The planar to cellular transition was suppressed in the presence of fluid flow. Also, convection was found to slightly increase the critical velocity for cellular to dendritic transition. © 2001 Elsevier Science B.V. All rights reserved.

## 1. Introduction

Because of the close relationship between the microstructure and properties of solidified materials, significant theoretical and experimental studies have been carried out to characterize solidification microstructures as a function of processing conditions [1,2]. However, rigorous tests of many theoretical models have been difficult since they are based on diffusive growth, whereas convection effects often dominate in most experimental studies carried out in bulk metallic samples. This is particularly true for cellular and dendritic growth at low velocities [3–8] and for layer

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structure formation in peritectic systems [9]. In the case of peritectic system, convection effects themselves have been shown to give rise to a new oscillating microstructure [10–14].

In order to test theoretical models developed for diffusive regime, convection-free growth conditions are required. Several different approaches have been taken to obtain diffusive growth. McCartney and Hunt [15] designed a ternary alloy of Al–Cu–Mg with composition adjusted so as to have negligible coefficient of density variation with composition. However, the presence of the third element complicates the solidification behavior and makes it difficult to obtain unambiguous interpretation. In addition, a radial temperature gradient is always present in Bridgman growth [12–14,16,17] that will give rise to fluid motion since there is no threshold for this condition [18,19]. Another technique that is often considered for suppressing convection is the use of a magnetic field during the solidification process. However, experimental studies using this technique have not given convection-free condition [20].

Since it is difficult to completely eliminate convection effects in experiments carried out terrestrially (i.e. under 1 g conditions) in bulk samples, it is important to quantitatively characterize how convection influences microstructure formation. Numerous theoretical models have been developed to characterize double diffusive convection that occurs in directional solidification for the case in which the rejected solute is lighter than the solvent [21–24]. In this paper, we shall examine systematically and quantitatively the effect of convection on microstructure formation for systems in which the rejected solute is heavier than the solvent. Specifically, our aim is to establish the experimental conditions under which diffusive growth could be obtained under terrestrial conditions. We have thus examined the effect of convection on vertically upward solidification of Al–4.0 wt% Cu alloys in a positive temperature gradient. The Al–Cu system was selected since convection effects have been shown to strongly influence the microstructure formation in this system [5,7,8]. Since the rejected solute is heavier than the solvent, and the temperature gradient is positive upwards, the stabilizing effects of thermal and concentration gradients on the density stratification ensures no fluid motion due to natural convection. However, fluid motion can occur due to the presence of radial temperature and concentration gradients. Several theoretical studies have established the presence of radial temperature gradients due to the geometry of the directional solidification process in which a thermally insulated region separates the hot and cold regions [12–14,16,17]. These radial gradients will always be present, and give rise to fluid motion [18,19]. Thus, convection effects on microstructure can be neglected only when the solute Peclet number, defined as the ratio of convective mass transfer rate to diffusive mass transfer rate, is smaller than one, or when the fluid velocity is much smaller than the externally imposed growth velocity. For a given system, the fluid velocity is characterized by a Rayleigh number (Ra) which, for a fixed horizontal temperature gradient ($G_H$), is given by

$$\mathrm{Ra_T} = \frac{g\beta_T G_H d^4}{\nu\alpha_\ell}, \tag{1}$$

in which $\nu$ is the kinematic viscosity, $\alpha_\ell$ is the thermal diffusivity of the liquid, and $\beta_T$ is the coefficient of density variation with temperature. The magnitude of the fluid motion will thus be controlled by the value of the Rayleigh number, so that one would need a Raleigh number that is sufficiently small to give fluid velocity that is much smaller than the externally imposed growth velocity. The Rayleigh number is proportional to $gd^4$ for a fixed horizontal temperature gradient, where $g$ is the gravity vector and $d$ is the diameter of the sample. Thus, convection effects can be reduced by reducing either $g$ or $d$. Microgravity environment with reduced $g$ offers an excellent opportunity to achieve diffusive growth unambiguously. However, it would be advantageous to carry out experiments terrestrially, not only in diffusive regime but also in convective regime where the effect of convection on microstructure development can be studied systematically and characterized quantitatively. Thus, the approach we have taken is to reduce the diameter of the sample, and study the microstructure selection as a function of sample diameter.

Most directional solidification studies in metallic systems are carried out in tubes that are between 3 and 7 mm in diameter, and we shall show that these diameters are still large enough to cause significant convective flows. To obtain solidification in the diffusive regime, fluid flow velocity must be significantly smaller than the growth rate of the interface. Since convective flow velocities decrease sharply as the diameter of the sample is reduced, a very fine sample diameter is required to minimize fluid flow effects on microstructure development. We shall first describe a numerical model to determine the sample size and growth conditions under which fluid flow effects become negligible in the Al–Cu system. We shall then describe an experimental technique for directionally solidifying very fine samples in which diffusive growth can occur. In this method, a bundle of very fine tubes, with diameters ranging from 1 to 0.2 mm are placed inside a larger tube of 5 or 6 mm diameter that is commonly used in the directional solidification experiment. All samples are then directionally solidified in the same experiment so that the microstructures formed in different diameter samples, with different fluid flow effects, can be quantitatively compared. Critical sample size below which diffusive growth conditions are established is then obtained when the microstructure or segregation profiles become independent of the sample size.

Numerical modeling, as well as experimental results, shows that convection effects in the Al–Cu system are negligible only when the diameter of the sample is 1.0 mm or less. By the comparison of microstructures in different size samples, it is shown that the stronger convection increases the cell/dendrite tip temperature and decreases the tip composition and primary spacing. In addition, convection increases the critical velocities for planar to cellular and cellular to dendritic transitions under fixed thermal gradient and alloy composition.

## 2. Numerical modeling

Numerical modeling of fluid motion is first developed to correlate the effect of convection with the sample diameter so that we can determine the size of the sample required for diffusive growth in the Al–Cu system. A detailed numerical model of the Bridgman crystal growth process has been developed for vertically upward solidification of a binary alloy of initial composition $C_0$ (in wt% solute) inside a two-dimensional rectangular cavity. The vertical walls are rigid solid walls that represent the three-zone thermal assembly and are impervious to mass flux. The three-zone assembly consists of a cold zone and a hot zone separated by an insulator. The system of coupled nonlinear equations describing the transport processes are written in terms of a coordinate frame moving with a constant speed equal to the imposed growth rate, $V_p$. The set of modeling equations includes heat conduction equations in the solid ampoule and the crystal, advection–diffusion equations for heat and solute transport in the melt, and the Boussinesq approximated Navier–Stokes equations for fluid flow due to thermal and solutal buoyancies. These equations are completely described by the thermal and the solutal Rayleigh numbers ($Ra_T$, $Ra_S$), the Peclet number based on growth rate (Pe), the partition coefficient ($k$), the Prandtl number (Pr), the Lewis number (Le) and the aspect ratio ($A$). The proper choice of the heat transfer coefficients was made through rigorous regression analysis of computed axial and radial temperature profile with measured experimental data in the directional solidification apparatus used in this study [25]. Alumina was used as an ampoule material for different size samples. Note that in the above variables only Rayleigh numbers are altered when the diameter of the sample is changed, so that the effect of sample diameter can be described by the variation in Rayleigh numbers which characterize the nature of fluid flow in the melt. The details of the numerical model are given in separate papers [12,14]. Here we shall mainly describe the results that are relevant to our experimental design.

The present calculations are carried out for an initially flat interface, and the curvature of the interface is obtained from the radial concentration variation and the temperature gradient. When the interface becomes curved due to the fluid flow effect, the curvature will be enhanced during

growth due to the lateral rejection of solute. Consequently, the results of interface shape in larger diameter samples, in which fluid flow is present, underestimates the actual curvature that will be present. However, this approximation would give a reliable result for conditions close to the onset of fluid flow where the curvature of the interface is small, and thus would allow us to determine the critical size of the sample below which diffusive growth can be obtained. In present calculations, we have also considered the horizontal temperature gradient to be fixed, and independent of the sample diameter. Experimentally, it is difficult to measure the radial temperature gradient in very thin samples. However, since the radial temperature gradient gives rise to the curvature of the interface, as a first approximation we assume that the radial temperature gradient will be proportional to the difference in interface temperature between the center and the wall. This temperature difference was estimated from the experimental results in the Al–4.0 wt% Cu system by measuring the deflection of the interface and using the thermal gradient value which does not vary significantly from the wall to the center. The results are shown in Fig. 1, and within the experimental scatter a constant value of the radial interface temperature gradient was observed.

A set of computations is carried out for directional solidification of Al–4.0 wt% Cu ($C_0 = 4$) grown at $V_p = 1\,\mu m/s$ in tubes of diameters ranging from 0.2 to 5 mm. We used two different nondimensional numbers to scale the relative strengths of convective and diffusive transports. The first number is a simple ratio of the convective velocity scale to the interface velocity ($U_c/V_P$). The maximum convective velocity in the melt within five diffusion lengths from the interface is taken as a measure for $U_c$. The second number is a solute Peclet number which is a ratio of the rate of convective mass transfer, $U_c/d$, to the rate of diffusive mass transfer, $D/V_p^2$, and is defined as

$$\mathrm{Pe} = \frac{U_c D}{d V_p^2},$$

where $D$ is the diffusivity and $d$ is the sample diameter. In Fig. 2, we plot both the nondimensional numbers as a function of sample diameter.

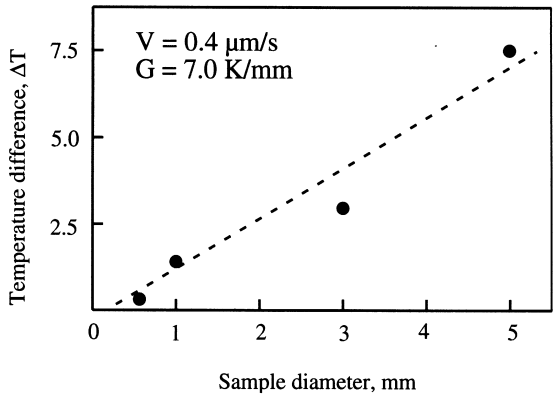


Fig. 1. Experimental measurements of the temperature difference at the interface between.the center and the side of the sample. Al–4.0 wt% Cu alloy.

Since, there is no sharp transition between diffusion and convection dominated regimes, we chose the criterion for the establishment of diffusive regime as ($U_c/V_P$) and $\mathrm{Pe} < 0.1$. It is observed from Fig. 2 that if the tube diameters are less than $\sim$1 mm, the convective transport is weak and molecular diffusion is the dominant mode of solute transport. For samples of diameter $\geqslant 3$ mm, the convection in the melt is quite significant and cannot be neglected.

Since the temperature near the ampoule wall is higher than in the center of the ampoule, the hotter liquid near the wall rises up and the liquid at the center moves downwards. Thus, along the interface, the flow causes the solute to move from the center towards the wall. The cellular convection in these samples thus causes lateral segregation of solute, both in the melt and along the solid–liquid interface. The effect of convection on the solute composition field (or segregation field) in the solidified sample is calculated for different sample diameters, and isoconcentrate contours are shown in Fig. 3. The four figures correspond to the solute fields in samples grown in tubes of diameters: 0.4, 0.6, 1.0 and 5.0 mm. Only a small length (1.2 cm) of initial transient is computed and presented due to the constraint on the computing facility. However, the general effects of increasing segregation, or departure from one- to two-dimensional diffusive segregation field, due to increasing convection in tubes of larger diameters is quite evident

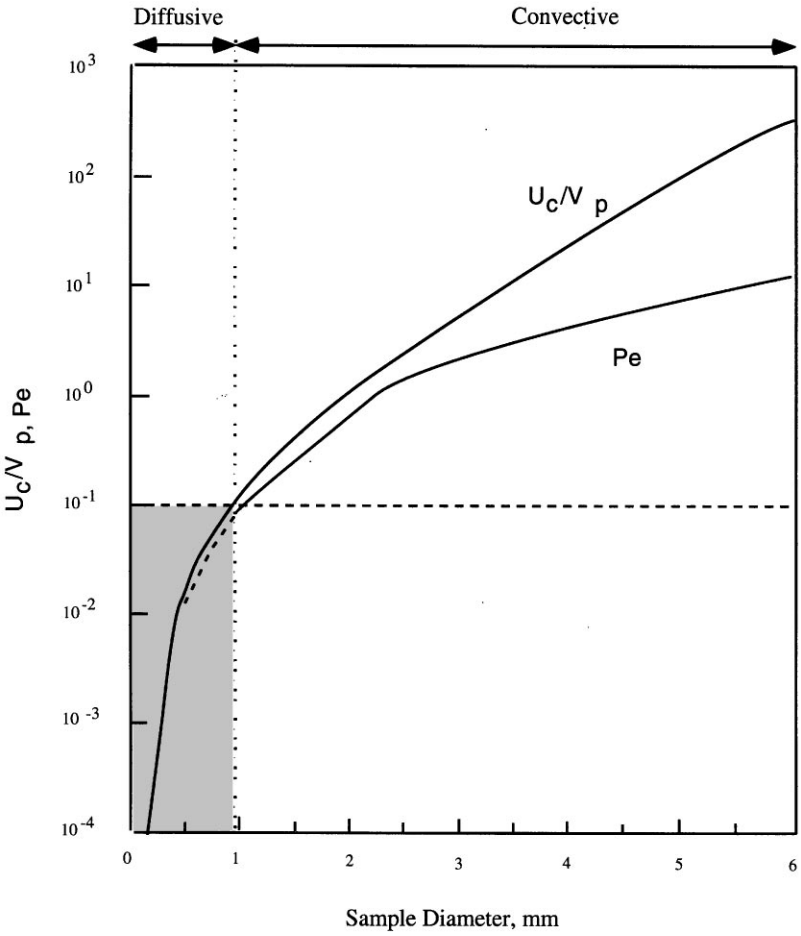


Fig. 2. The ratio of the maximum melt velocity and the crystal growth velocity, $U_c/V_p$, and the ratio of convective rate to diffusive rate of solute transport ($U_c D/dV_p^2$) as a function of sample diameter in an alumina ampoule in the Al–Cu system. Note that diffusive growth conditions require sample diameters smaller than 1 mm.

in these plots. These calculations show that although convection is present in all the samples, the concentration field is not significantly affected by the fluid flow if the sample diameter is less than 1 mm. Consequently, the microstructures would exhibit diffusion-dominated growth in samples of diameters less than 1.0 mm. Note that this conclusion is valid only for Al–Cu alloys in alumina tubes with thermal profiles similar to those in our directional solidification apparatus.

The above calculations assume that each thin sample is directionally solidified individually in a thermal assembly. In the directional solidification set-up, a significant radial temperature gradient can be present due to the discontinuity in the boundary conditions along the outer wall of the ampoule, i.e. requirement of zero flux in the adiabatic zone. In the experimental setup described below, several samples are placed in the melt so that each thin sample is surrounded by a liquid which smoothes out the boundary condition, thereby avoiding the discontinuity which will reduce the curvature of the isotherms. Thus, our numerical model over-predicts segregation for the specific experimental design used in this study. In addition, our calculations show that the isotherms in the directional solidification have larger curvature near the edges of the adiabatic zone than at the center. Thus, if the assembly is designed such that the tube in which the alloy is solidified is

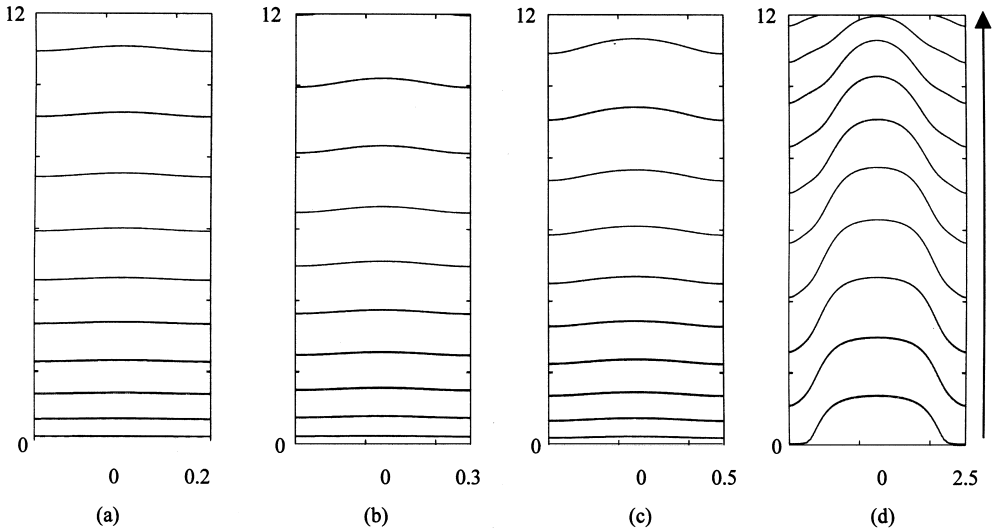


Fig. 3. Isoconcentration contours in the solidified crystal in alumina ampoules of diameter: (a) 0.4 mm (b) 0.6 mm (c) 1.0 mm (d) 5.0 mm. The distances are in mm and the arrow shows the growth direction.

placed in the center region of a larger tube which also contains the alloy, isotherms in the actual sample will be flatter so the fluid motion will be reduced. A concentric tube assembly is thus suggested to reduce convection.

## 3. Experimental studies

In order to obtain diffusive growth, the results of the numerical analysis show that samples of diameters less than 1.0 mm are required for the directional solidification of Al–Cu alloys at low velocities. The experimental setup to directionally solidify very fine samples is difficult. However, we have designed a technique in which conventional solidification equipment can be used to solidify thin samples. We shall now describe this technique and show the effects of convection on microstructure development in Al–Cu alloys.

### *3.1. Experimental procedure*

The directional solidification equipment used consisted of a thermal gradient assembly and a translation system. The thermal assembly contained a furnace and a cooling chamber, which were separated by an insulating block. A tube of inner diameter (ID) 5.5 mm which contained the alloy was next inserted in the thermal assembly. In some experiments, a single-crystal seed with ⟨0 0 1⟩ direction along the axial growth direction was used to ensure that cells and dendrites grow along the axial direction. In the conventional technique, the sample is held stationary and directional solidification is carried out by moving the thermal assembly upwards. We have modified the sample by placing a cluster of smaller diameter tubes inside the 5 mm alumina tube, as we shall now describe.

The sample assembly was first designed by making a cluster of several smaller diameter tubes that were next inserted into the molten metal in the 5.5 mm ID tube. The arrangement of tubes is shown in the longitudinal and transverse sections of the sample tube assembly in Fig. 4. In some experiments, one of the finer tubes contained a thermocouple to characterize the thermal profile during growth (Fig. 4b). Diameters of smaller tubes in the cluster ranged from 0.2 to 1.0 mm. The liquid region outside the small tubes was about 3.2 mm (Fig. 4c). The finer tube assembly, before placing it in the molten alloy, was held together by an alumina cement, which was cured for 1 h at 500°C to remove volatile materials. A sufficient space between the tubes was maintained by using

cement at predetermined positions which held the tubes in a proper vertical direction and which allowed the liquid alloy to move through the spaces between the tubes. Initially the alloy was melted in 5.0 mm alumina tube under vacuum, and then the tube assembly was inserted into the molten alloy. For tubes of diameters 1–3 mm, a small positive pressure of argon was used to ensure that the liquid alloy fills the inner tubes. The filled tubes were then held for about 40 min to stabilize the thermal field.

The samples were initially solidified rapidly for 50 s and then equilibrated so that the solid–liquid interfaces were inside all tubes, and the location of the interface in all tubes was the same, as shown by the position B in Fig. 4a. This fixed initial reference point is important since it will allow us to determine the relative locations of the quenched interface in different tubes, thereby showing the effect of convection on the relative interface position and interface temperature. Directional solidification was then carried out by moving the thermal assembly over a predetermined distance, after which the sample assembly was quenched. In this technique, all samples with different diameters were directionally solidified under identical conditions so that the differences in microstructure could be related to the size of the tube, and thus to the extent of convection. After the directional solidification run, all samples were removed, polished, etched and their microstructures characterized.

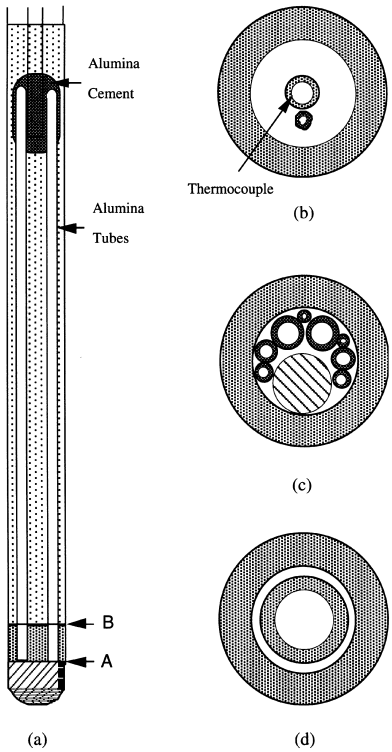


Fig. 4. Arrangements of sample assembly in which several fine tubes of different diameters are placed inside a standard 5.0 mm alumina tube. (a) Longitudinal section. (b) Transverse section showing the presence of a thermocouple to characterize the thermal profile. (c) Transverse section of one arrangement of the tube assembly without the thermocouple. (d) Transverse sections of two concentric tubes giving a narrow annular region. Longitudinal section also shows the location, A, of the initial interface when the fine tubes are placed in 5 mm. ID alumina tube and the position of the interface, B, when directional solidification was initiated after initial rapid solidification.

### 3.2. Experimental results

A set of experiments was carried out in the Al–4.0 wt% Cu system to examine the effect of convection on cellular growth and on the planar to cellular, and cellular to dendritic, transitions. Experiments were carried out over a range of velocities, 0.2–20.0 μm/s, and the temperature gradient was varied from $G = 5.0-11.0 \pm 0.5$ K/mm. The lower and upper limits of velocity were chosen to span the planar, cellular and dendritic microstructural regimes. The directional solidification experiments were carried out in two sample arrangements, the fine sample assembly consisting of thin tubes of diameters 0.2, 0.4, 0.6 and 1.0 mm inside a 5.5 mm tube and the bulk sample in 5.5 mm tube only. The microstructures in the large open region in the fine sample assembly were found to be similar to those observed in bulk samples of 5.5 mm diameter.

The variations in microstructures with velocity and sample diameter are shown in Figs. 5–8. The samples in Fig. 5 were directionally solidified at low velocities where planar front was observed. In a larger (5.0 mm) diameter sample, the interface shows a significant macroscopic curvature due to radial concentration variation caused by fluid flow, as predicted by the model (Fig. 3). In this

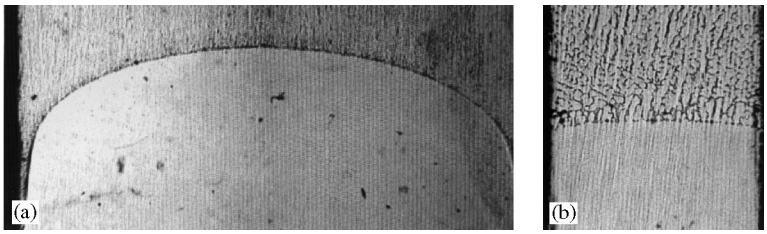


Fig. 5. Experimental results in the Al–4.0 wt% Cu system for $V_p = 0.4\,\mu m/s$ and $G = 7.0\,K/mm$ in samples of diameters (a) 6.0, and 0.6 mm. The curvature of the interface in larger diameter sample is due to the lateral solute segregation caused by convection.

sample, no smaller wavelength instability is observed. In thinner samples of diameters 0.6 mm the interface is observed to have a much smaller curvature.

The microstructures at a higher velocity are shown in Fig. 6, which shows that the interface is curved, but has not developed instabilities in a larger 5.5 mm diameter sample. In thin samples of 1.0 mm diameter and less, cellular structures were observed. The microstructure in 0.6 mm diameter samples was found to consist of a single cell. A comparison with the results in Fig. 5, where a flatter interface was observed in thin samples if the microstructure were to be planar, shows that the microstructure in thin sample in Fig. 6 is indeed cellular. In order to clearly establish this conclusion, experiments were carried out in 1.0 mm diameter sample that was placed in a 5.5 mm diameter tube. The transverse section of the sample is shown in Fig. 7a. It is clear that the microstructure inside the 1.0 mm tube is cellular, whereas a planar front solidification is present outside the 1.0 mm tube where significant convection effects are present. Similar results were observed in Sn–0.8 wt% Cd system in which the rejected solute is heavier as in the Al–Cu system (Fig. 7b). Thus, in thin samples where diffusive growth occurs, a cellular structure is stable, whereas in a larger sample a planar interface is stabilized due to the effects of convection. Note that in presence of convection, the solute build up is slower so that it would require a significantly longer solidification distance at low velocities to conclude that the planar interface is indeed stable under these growth conditions.

Experiments near the cell–dendrite transition velocity (Fig. 8), show that cellular/dendritic

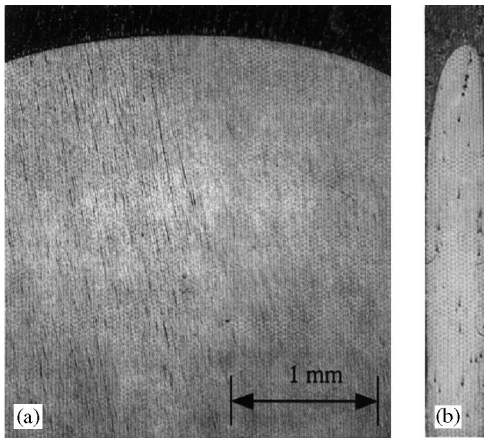


Fig. 6. Experimental results in the Al–4.0 wt% Cu system for $V_p = 1.3\,\mu m/s$ and $G = 9.5\,K/mm$ in different diameter samples. (a) A planar, curved front is observed in 3.2 mm diameter sample, and (b) a cellular structure is observed in a thin sample of diameter 0.6 mm.

structures are present in 5.5 mm diameter sample, whereas well-developed dendritic structures are formed in smaller diameter samples. This result indicates that convection effects also tend to stabilize the cell to dendrite transition.

All experimental results on microstructural features are summarized in Fig. 9. Fig. 9a and b shows all results in samples of diameter $\geqslant 3$ and $\leqslant 1.0$ mm, respectively. No planar front was observed in thinner samples, whereas planar front was present in bulk samples for the same solidification lengths.

In order to examine the solute redistribution, the low-velocity samples were analyzed for axial composition variation at the center of the interface. The results for 0.4 and 3.0 mm diameter sample are shown in Fig. 10, which clearly show

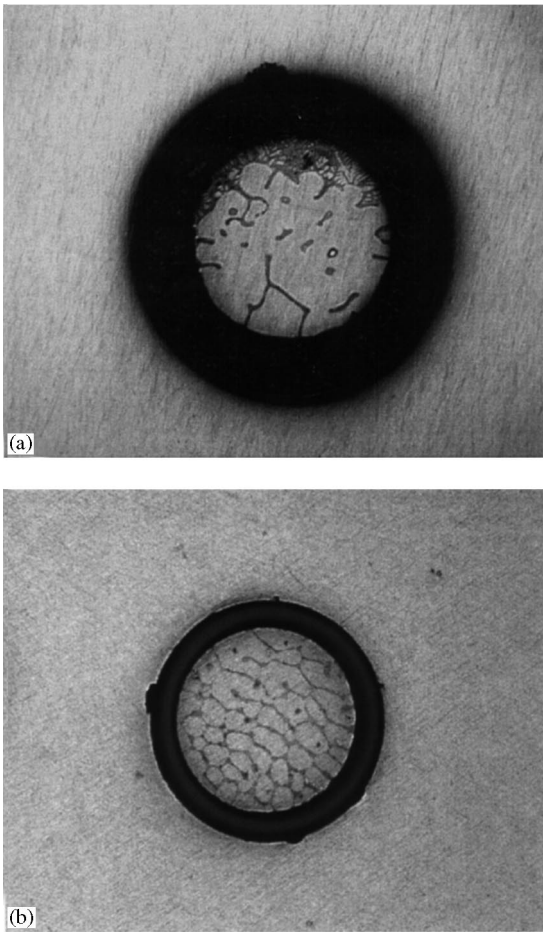


Fig. 7. Microstructures formed inside and outside a sample tube of 1.0 mm in diameter. In the 1.0 mm sample a cellular microstructure is observed, whereas outside the sample, where convection effects are present, a planar front solidification takes place. (a) Al–4 wt% Cu, $V = 0.8\,\mu m/s$, $G = 7.5\,K/mm$, and (b) Sn–0.8 wt% Cd, $V_p = 8.0\,\mu m/s$, $G = 20\,K/mm$.

that the solute build up and the composition in the solid are higher in 0.4 mm sample in which diffusive growth occurs. Composition profiles in solid were also measured in an another experimental run with identical $V_p$ and $G$. The results are shown in Fig. 11. It is seen that the composition profiles are identical for sample diameters of 1.0 mm and less, whereas a significantly lower composition is found in bulk samples. This result also confirms that diffusive growth is obtained in samples of 1 mm diameter or less. It should be noted that the increase in composition near the interface is an artifact of the slower quenching rate in this sample. During quenching the interface advances at a higher rate before it becomes unstable, and this increase in solidification rate causes increase in the solute content in the solid.

Microsegregation profiles were also measured for cellular microstructures, for samples solidified at $V_p = 4.0\,\mu m/s$. These composition profiles were measured on transverse sections behind the cell tip location. The results for different diameter samples are compared in Fig. 12. These results again show that the composition values are smaller for samples of larger diameter sizes, and the composition profiles are identical for samples of diameter $\leqslant 1.0$ mm. These experimental results show that the fluid flow effects on cellular microstructure is also negligible only when the sample diameter is less than 1.0 mm for directional solidification of Al–Cu alloys at low velocities. These results are consistent with the results of the numerical modeling.

## 4. Discussion

### 4.1. Sample assembly design

A new design of the sample assembly is described which allows one to directionally solidify several thin samples simultaneously in one experiment. Since the solidification initiates at a fixed location in all samples, the comparison of the quenched interface allows one to accurately characterize the relative effect of sample diameter on the position of the interface, and thus on the interface temperature and composition. Since the convection is characterized by the Rayleigh number, which is a function of the tube diameter, it is possible to quantitatively relate microstructure features with the Rayleigh number. In addition, this technique allows one to determine the critical Rayleigh number for which the microstructural features approach those predicted by the diffusive model. Our experimental results in the Al–Cu system show that a sample diameter of 1 mm or smaller is required to obtain diffusive growth. Using this critical experimental value of the Rayleigh number, a diffusive growth could be

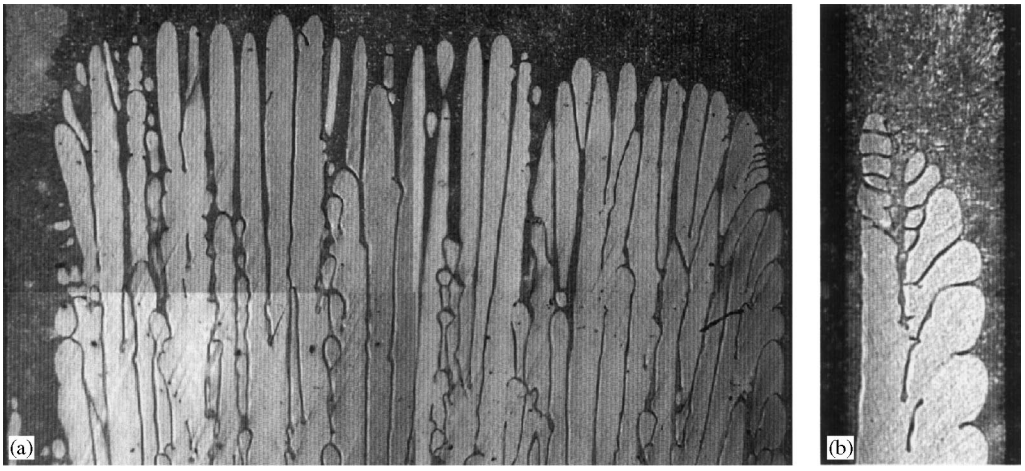


Fig. 8. Experimental results in the Al–4.0 wt% Cu system for $V = 5.0\,\mu m/s$ and $G = 7.0\,K/mm$ in different diameter samples. (a) A mainly cellular front is observed in 5.5 mm diameter space, and (b) dendritic structures are observed in a thin tube of diameter 0.6 mm.

achieved in a 6 mm diameter tube, if experiments were carried out under microgravity levels of $10^{-3}$ g or less. Similarly, one can determine the microgravity level required for diffusive growth in samples of different diameters. Alternately, if the microgravity level is known, the maximum size of the sample that would give diffusive growth can be established for a given system.

One of the important aspects of the design is that thin samples, in which directional solidification is carried out, are surrounded by a liquid alloy with the same thermal properties. Consequently, the heat transfer would be quite rapid. It has been well documented in the literature that in the directional solidification setup, with an adiabatic zone between the hot and cold zones, a significant radial temperature gradient can be present due to mismatch of boundary conditions along the outer wall of the ampoule [14,16,17]. In the modified sample assembly the axial thermal profile in the liquid between tubes is smoothed out so that the curvature of isotherms is reduced. Furthermore, the isotherms in the directional solidification have larger curvature near the edges of the adiabatic zone than at the center. Thus, if the assembly is designed such that the tube in which the alloy is solidified is placed in the center of a larger tube which also contains the alloy, isotherms in the actual sample will be flatter, thereby reducing fluid motion in the systems due to radial thermal gradient. The arrangement in this case is shown in Fig. 4d. Such a double-tube design is expected to significantly reduce the convection effects in the inner sample tube sample, and it is recommended for directional solidification studies. Note that this arrangement would also reduce fluid flow in the concentric region of the two tubes, and would give diffusive growth if the gap between the two tubes were small.

### 4.2. Microstructural observations

In order to examine the reduced effect of convection on microstructures, we have examined two phenomena in which convection effects have been known to significantly alter microstructural characteristics. Convection effects have been shown to be dominant in the formation of cellular structures at low velocities [5,26,27], and on the planar to cellular and cellular to dendrite transitions. We have thus carried out experiments in the Al–Cu system to examine the changes in microstructures as the diameter of the tube is reduced.

Several interesting features of microstructures have been observed in the experiments in the Al–Cu system. Figs. 6 and 7 show that a planar front is present in the bulk sample $\geqslant 3.0$ mm diameter, whereas in a thin tube $\leqslant 1.0$ mm diameter, a cellular microstructure is observed under identical growth conditions and solidification distance. This

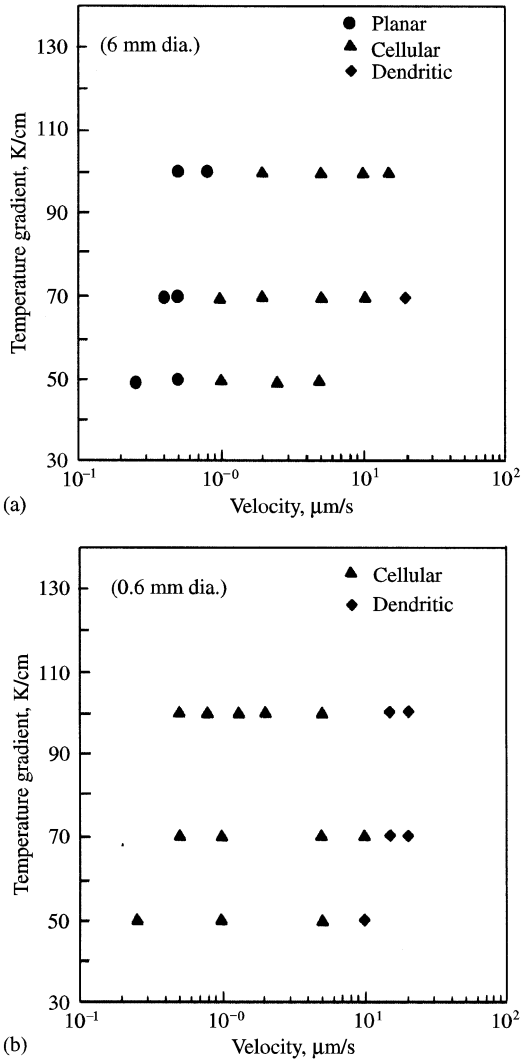


Fig. 9. The planar, cellular and dendritic regimes as a function of $G$ and $V$. Experimental results in: (a) sample diameters $\geqslant 3.0$ mm where significant convection effects were present, and (b) sample diameters $\leqslant 1.0$ mm in which diffusive growth was present. The planar interface points are for finite length of solidification for which steady state growth conditions were not achieved.

results show that convection effects in a larger diameter sample stabilized the planar interface since convection reduces the solute gradient at the interface in the liquid.

Several qualitative observations were also made. The location of the tip position in thinner samples showed a small decrease in tip temperature as the tube diameter became smaller. The interface in the bulk sample was at a much higher temperature. This increase in tip temperature is consistent with the observation of decrease in tip composition as convection effects become important. Also, the cellular spacing was found to increase as the diameter was reduced. Although these results give some insight into the effect of convection, they could not be analyzed quantitatively since the width of the sample significantly influences the development of a cellular array. When the sample diameter is slightly larger than the cellular spacing in an array, the cell tip keeps splitting and steady-state cells are not formed. Thus, quantitative examination of cell tip temperature and composition will require a more detailed study with a sample diameter that is close to the characteristic cellular spacing for a given growth condition so that only one (or seven) steady-state cell forms in a given sample. In addition, for dendritic growth, an accurate alignment of the growth orientation is required for the dendrite to be stable along the entire length of a thin sample.

Experimental results at $V_p = 5.0\,\mu m/s$ and $G = 7.0\,K/mm$ show the formation of a cellular/dendritic structure in bulk sample, whereas a dendritic structure is observed in a thin sample of 0.6 mm diameter. This observation again shows the stabilizing effect of convection on the cellular structure, so that the presence of convection increases the velocity of cell–dendrite transition. This increase in velocity, however, is much smaller since the effect of convection is reduced at higher velocities. No accurate model of cell–dendrite transition has yet been developed, so that the effect of convection on this transition can only be explained qualitatively due to the reduction in concentration gradient near the cell tip.

In the study of cellular/dendritic growth in directional solidification, two important aspects need to be considered, the stability of a single cell or a dendrite and the stability of an array. In experiments carried out in bulk samples these two effects cannot be isolated. However, if one can form a single cell or a single dendrite in very small diameter tube, then the diffusive transport model can be tested. To achieve a single cell or dendrite that can grow in a steady-state manner, specific

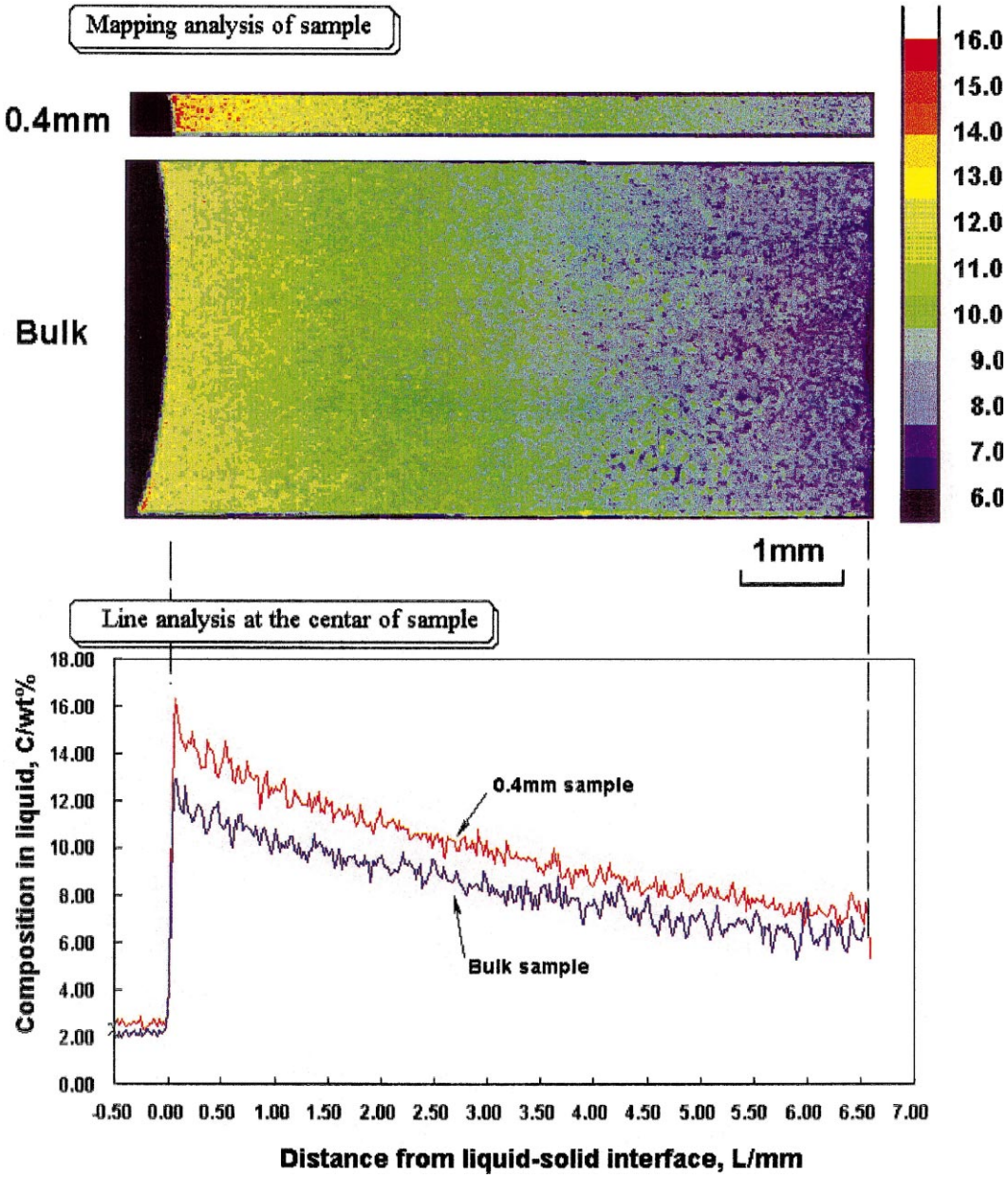


Fig. 10. Concentration mapping and concentration profiles in the solid and in the quenched liquid for planar interface growth in 0.6 and 3.0 mm sample diameters. $V_p = 0.4\,\mu m/s$, $G = 11.0\,K/mm$.

growth conditions are required for a given sample diameter size such that the diameter is smaller than the spacing that would cause tip splitting. In addition, the preferred orientation of the cell or dendrite must be along the axis of the sample. If the preferred growth direction is not aligned with the axis, asymmetric cell or dendrite will form in a constrained tube size that may not give a steady-state growth. A detailed experimental study and the analysis of single cells and single dendrites are now being carried out to test the diffusive growth model. Note that, it is not possible to examine the

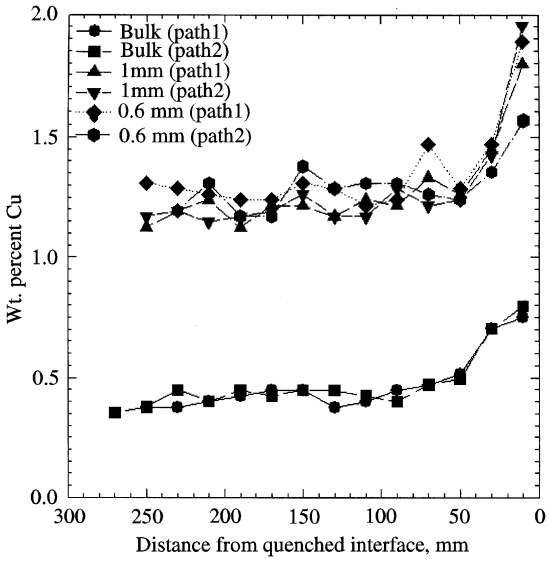


Fig. 11. Concentration profiles in the quenched solid for a planar interface growth in different diameter samples. Growth conditions: $V = 0.4\,\mu m/s$, $G = 11.0\,K/mm$. Two sets of data are taken in each sample at different locations for ensuring the reproducibility of the results. The increase in concentration close to the interface (~50 μm distance) is due to the quenching effect which caused the planar interface to advance faster.

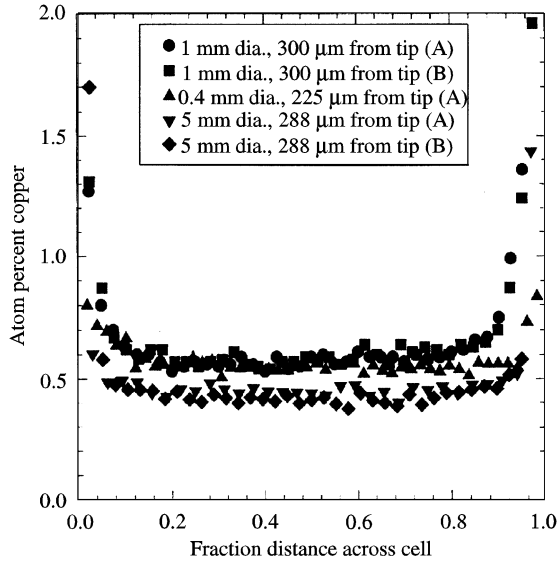


Fig. 12. Radial composition profiles, from transverse sections behind the cell tip, for sample diameters in directionally solidified Al–4.0 wt% Cu. $V_p = 4.0\,\mu m/s$, and $G = 11.0\,K/mm$.

cellular/dendritic array stability in diffusive regime by this technique since the presence of several cells or dendrites requires a larger sample diameter in which convection effects become dominant. Primary dendrite spacing under diffusive growth conditions can, however, be examined in 1.0 mm diameter sample grown under very high growth rates where the primary spacing is of the order of 50 μm or less so that sufficient number of dendrites are present for a meaningful analysis.

### 4.3. Applications

Although the experimental method has an advantage in that diffusive regime can be accessed, the size of the sample can significantly alter the microstructure. Thus, the scale of microstructure to be examined should be significantly smaller than the sample diameter. This method will thus be very advantageous in studying diffusive growth for planar front, and it can also be used to study diffusive growth in eutectic and monotectic systems since the characteristic spacing are generally very small so that the finite constraint of the sample will not influence their microstructural characteristics.

Recently, Park and Trivedi [10] have used this technique to show the effect of convection on the formation of an oscillating structure in the two-phase region of peritectic system. By comparing the microstructures in thin and bulk samples, they have shown that the convection effects are responsible for the formation of a novel oscillatory tree-like structure. Also, a discrete diffusive band formation was obtained for the first time in thin samples, within a small range of composition in the hypoperitectic region, where the diffusive model predicts the formation of bands that is controlled by the nucleation undercooling of the two phases [28].

This technique can also be used advantageously to examine the orientation dependence of kinetic effects in a pure material by forming several grains in the solid before the finer tube assembly is introduced, i.e. location A in Fig. 4a. A rapid solidification over a short distance will allow different orientations to enter different tubes, and will form the solid–liquid interface at the same horizontal level when the sample is equilibrated

with the liquid. Directional solidification will now occur in all tubes from the same reference level, so that when the liquid is quenched after directional solidification, the relative locations of quenched interfaces in these tubes will give the difference in interface temperatures, and thus the ratios of kinetic coefficients. The orientation of each sample can be determined by X-ray analysis after the sample is solidified and removed from the tubes. Since the kinetic effect is small, the direct measurements of differences in interface temperature will give reliable values of relative kinetic coefficients.

## 5. Conclusions

A numerical model is examined to obtain conditions under which diffusive growth can occur. Based on this analysis, a new experimental technique is described which allows a systematic study of the effect of convection on solidification microstructures in samples of different diameters. It has several advantages. (1) Since initial solidification interfaces of all the samples are at the same isotherm, the location of the quenched interface gives a relative information on the effect of convection on interface temperature and composition. Since one measures the difference in positions, a small deviation can be accurately characterized. (2) All samples with different diameters are growing in one tube under precisely identical conditions so that their results can be reliably compared. (3) Thin samples are surrounded by liquid alloy so that heat transfer is very rapid. In addition, the thermal profiles near the interface can be produced with a smaller radial temperature gradient since the axial thermal field in the outside liquid region is smoothed out so that zero flux condition at the walls of the finer tubes in the adiabatic zone is not required. Such a double tube design, shown in Fig. 6d, is recommended to decrease convection effects. (4) By observing microstructures in tubes with decreasing diameters, or decreasing Rayleigh number, one can identify the effect of convection on microstructure development, and relate the microstructural characteristics such as interface temperature, or composition, to the sample diameter or Rayleigh number. (5) It allows one to determine the size of the tube required for diffusive growth so that diffusive growth can be accomplished for comparison with theoretical models. The present technique also allows one to determine the sample diameter for which convection effects will be negligible under known microgravity environment. For example, since convection effects were negligible only for sample diameters of 1 mm or less in Al–4.0 wt% Cu alloy, one can predict that a sample of 1 cm diameter would solidify in a diffusive regime only under microgravity level of $10^{-4}$ g or better for the same value of the radial temperature gradient.

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